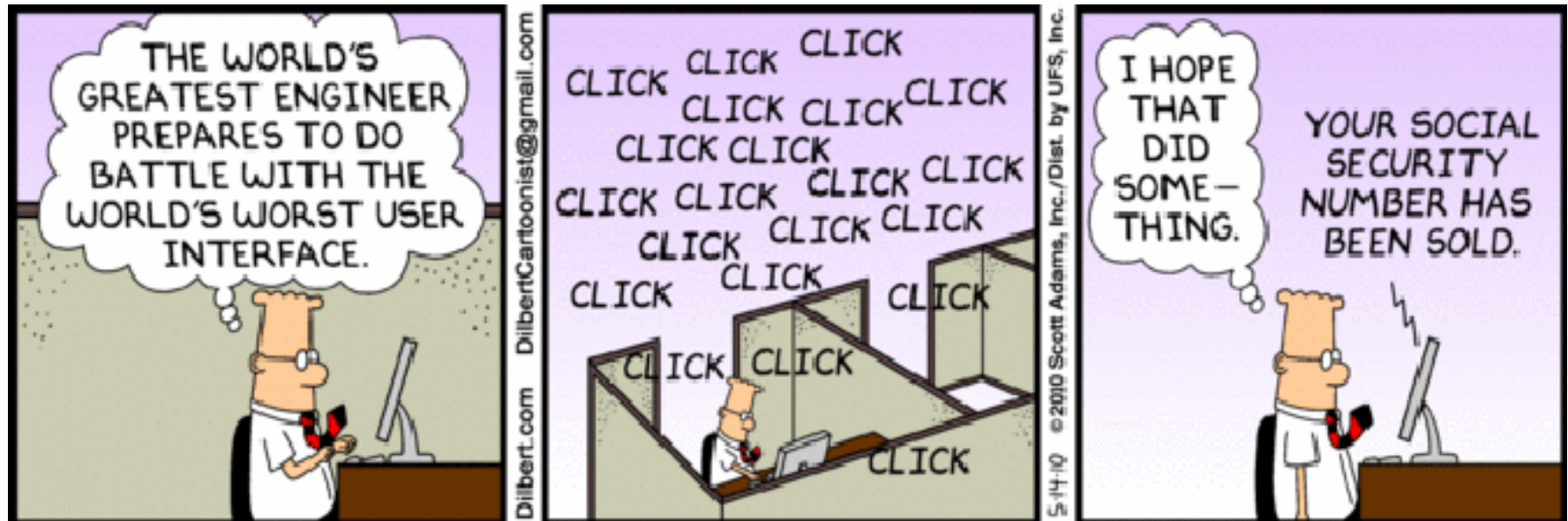


UI TESTING

WHY?



WHAT?

- functionality
- compatibility
- performance
- security
- installation of the app
- usability

FUNCTIONALITY TESTING

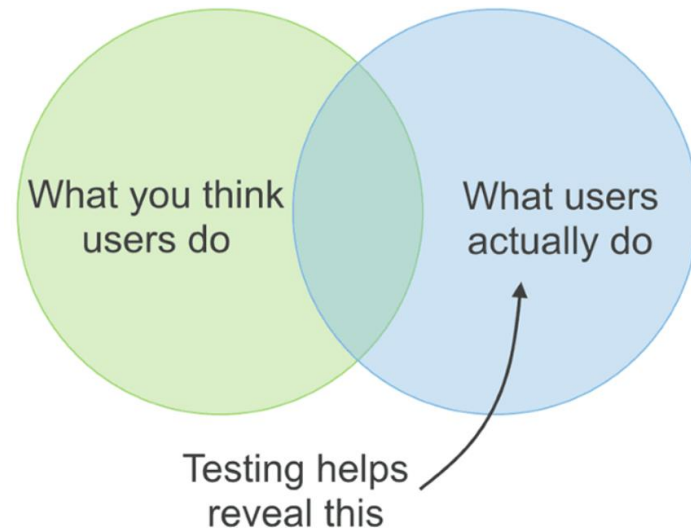
- functions:
 - can all tasks be performed?
 - do input fields accept the correct type of input?
 - is all input correctly checked?
- layout / styles:
 - were guidelines followed in whole app?
 - are font type and size consistent throughout the app?
 - are all the icons in the same style?
- display :
 - are the GUI elements correctly displayed on different devices and screens?
 - are the images correctly displayed?

FUNCTIONALITY TESTING

- Manuel:
 - User tests GUI
 - repetitive
- Automatic:
 - Companies that perform testing for a fee
 - Test writing tools :
 - Open source: Selenium, QTP, Cucumber
 - Android Studio
 - Visual Studio

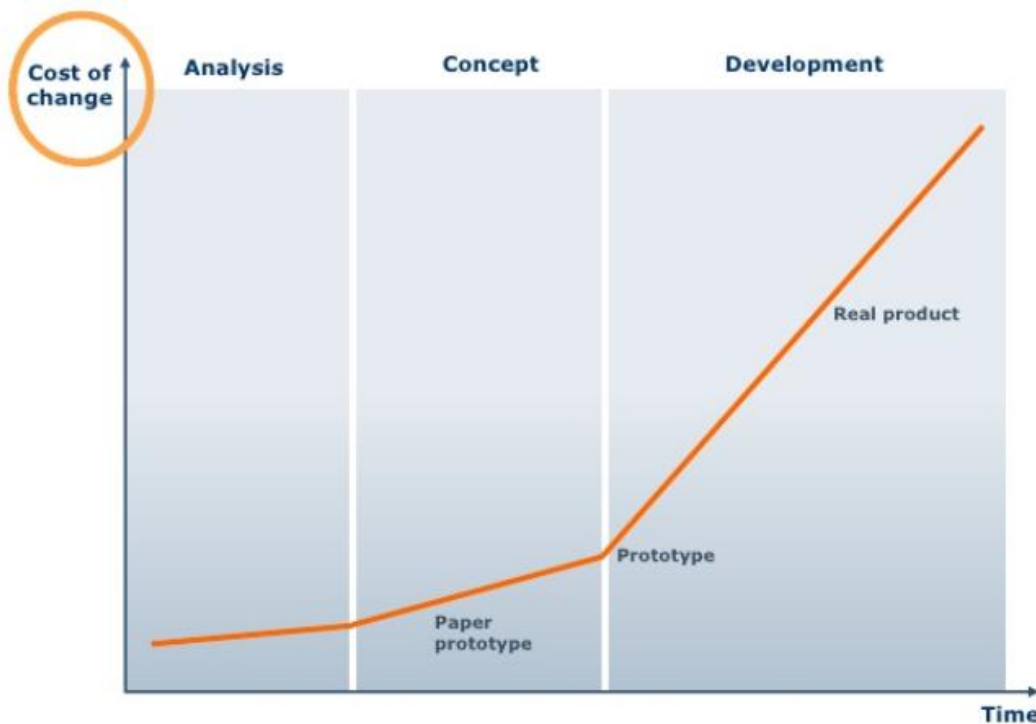
USABILITY TESTING

- If you want a good app, you need to test it
 - Not everyone thinks as you do
 - Everyone is different



USABILITY TESTING: WHEN?

- Testing with just one user early in the project is better than asking 50 users to test at the end



PROJECT START: ANALYSES

- test existing applications
- redesign:
 - test the current version of the application
 - study helpdesk data

⇒ What should be improved?

⇒ What can stay as is?

CONCEPT PHASE

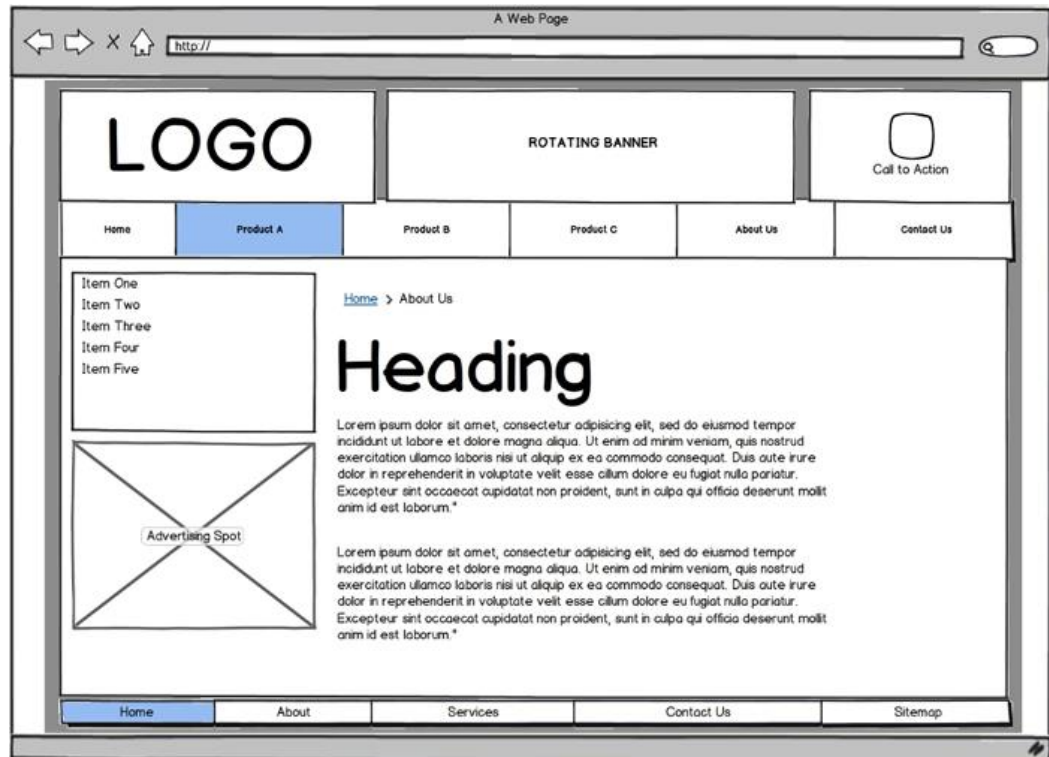
- Design information architecture
 - Card sorting:
 - let users divide cards with content/tasks from the app into logical groups
 - ask a name for each group
- ⇒ logical grouping?
- ⇒ good structure?
- ⇒ good, clear terminology?

CONCEPT PHASE

- Wireframe testing

⇒ good navigation scheme?

⇒ all elements present?



APP DEVELOPMENT

- Testing of more detailed wireframes, prototypes and different product builds

⇒ fine-tuning

⇒ validation

Prototype

From **Sketch** to **High-Fidelity** mock-ups to
Interactive Prototypes



TEST METHODS

Usability Testing Methods

Moderated testing

In-Person Usability Testing



Tester is directly observed and interacted with during the testing session

Monitored Remote Usability Testing



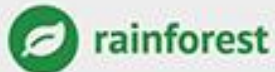
Tester completes actions remotely but is observed or engaged remotely

Unmoderated testing

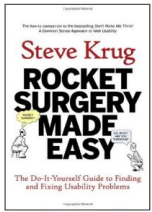
Unmonitored Remote Usability Testing



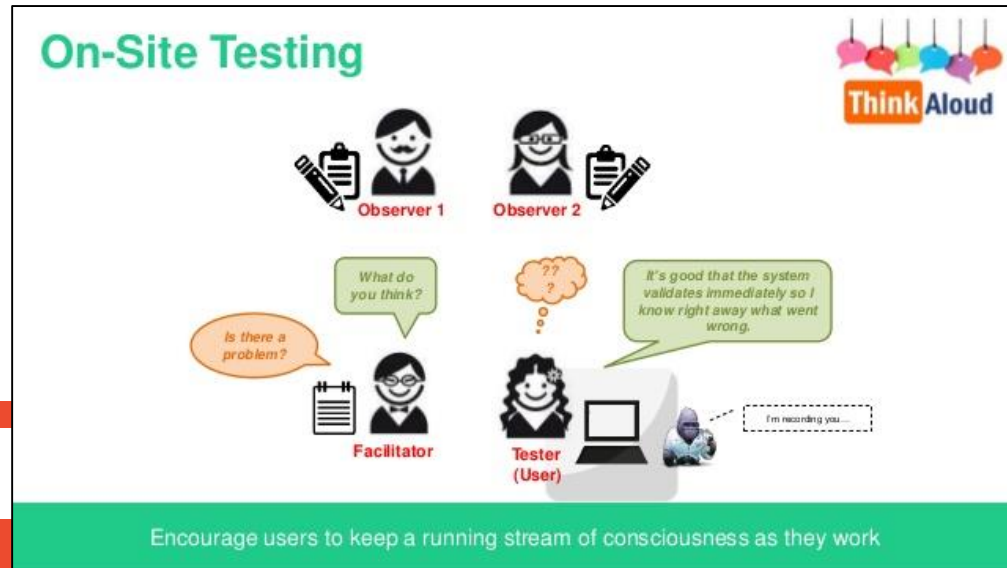
Tester completes actions and results are reviewed at a later time



MODERATED TESTING



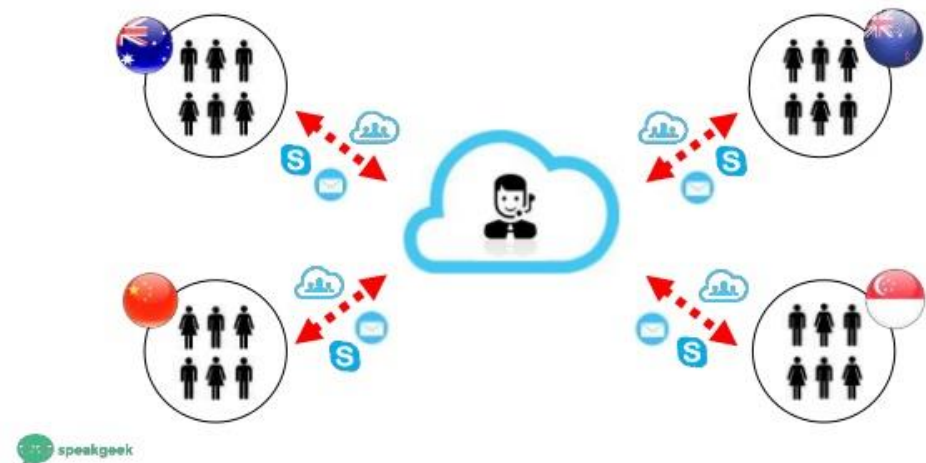
- Let users perform tasks while you observe them
 - around 90 minutes
 - small group (+/-5 users)
 - preferably from the target group
 - in a quiet room computer with microphone, webcam, possibly screen capture tools and internet (=> streaming to someone who follows along)
 - results are **qualitative**
=> you can understand why the user behaves as he does.



UNMODERATED TESTING

- Record voice, screen and click actions of users performing tasks
 - test a large group of users
 - Online
 - special software programs
 - results are **quantitative** => statistics

Remote Testing

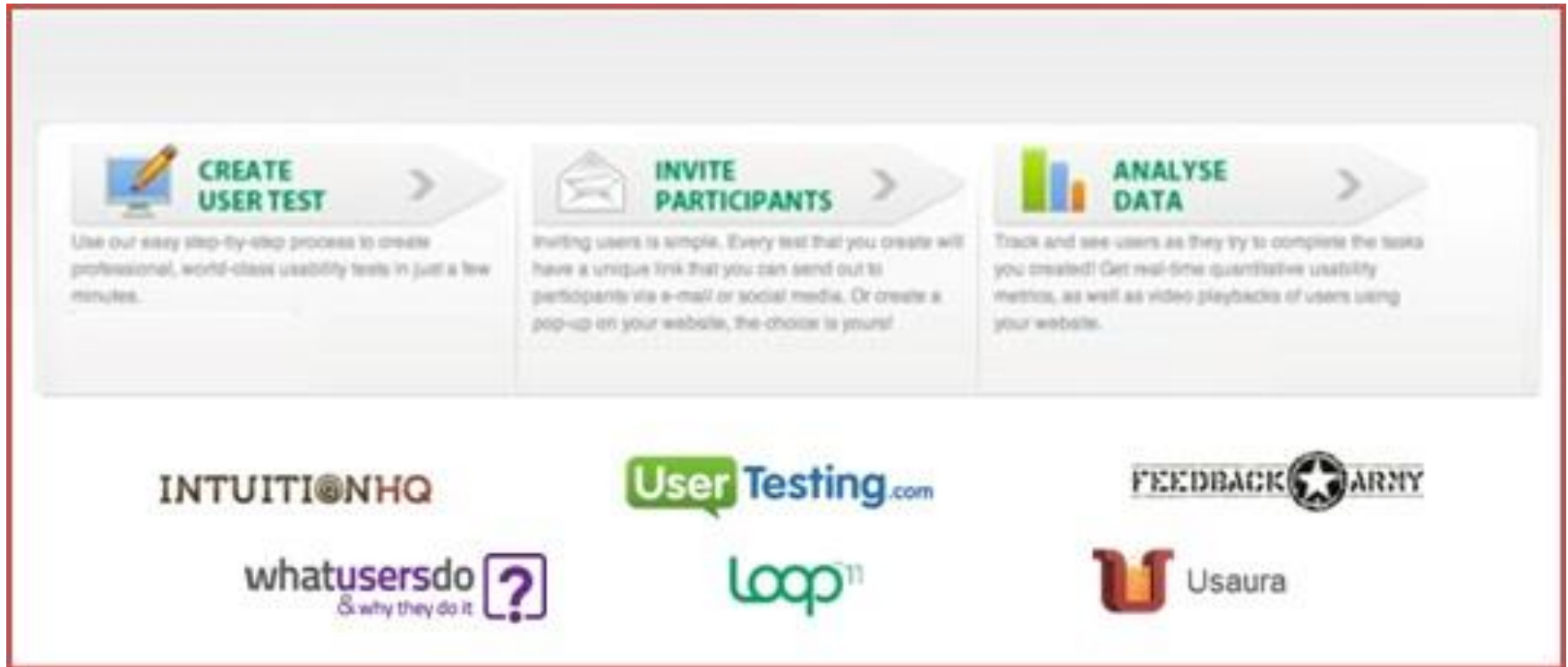


LEARNABILITY, EFFICIENCY

- let users perform a number of tasks
- choose tasks that are important
- Unmoderated testing -> measure:
 - Time needed to complete the tasks
 - Number of steps needed to complete the tasks
 - Number of tasks that could be performed correctly
- Moderated testing -> observe:
 - What does the user look at first?
 - How does the user behave?
 - Why does he sigh, laugh, ...Make user think out loud

TEST TOOLS

- User testing (execution of tasks by remote users): IntuitionHQ



TEST TOOLS

- Mousetracking: ClickTale, ClickDensity



TEST TOOLS

- Eyetracking:



MEMORABILITY / ERROR

- Error:
 - How many mistakes were made?
 - What was the time needed to correct an error?
 - How many fatal mistakes were made?
- Memorability:
 - Make test subjects come back after a longer period and perform the same tasks
 - Measure the same parameters

TEST TOOLS

- Memory test: Fivesecondtest, Clue
 - Displays web page for a few seconds
 - Then ask the user to answer questions about the web page

SATISFACTION

- surveys:
 - How difficult/easy were the tasks?
 - Stress level during execution of the tasks?
 - Frustration level?

!! Very subjective

Kate Meyer:

visually appealing design

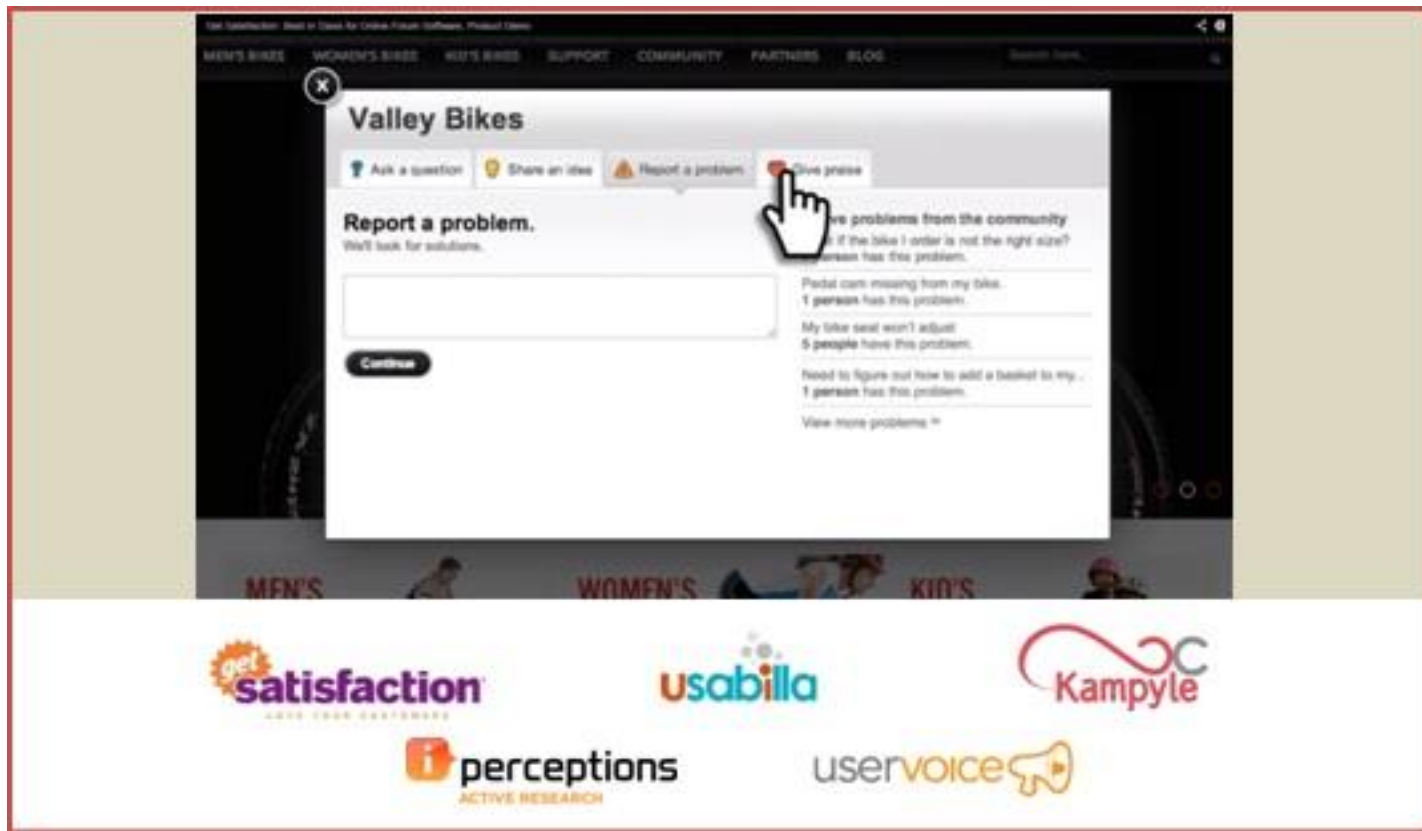
=> users are more tolerant

=> shortcomings are missed

=> observe the user, listen to feedback

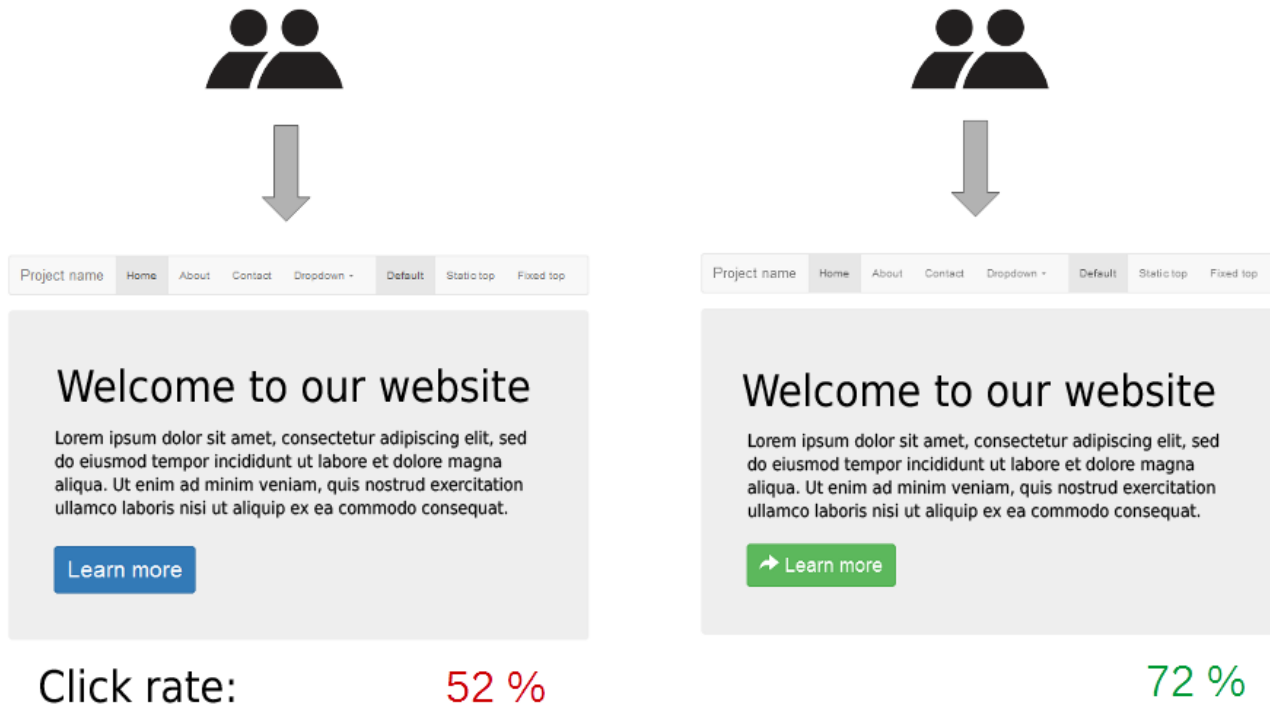
TEST TOOLS

- User feedback/ surveys: UserVoice, 4Q



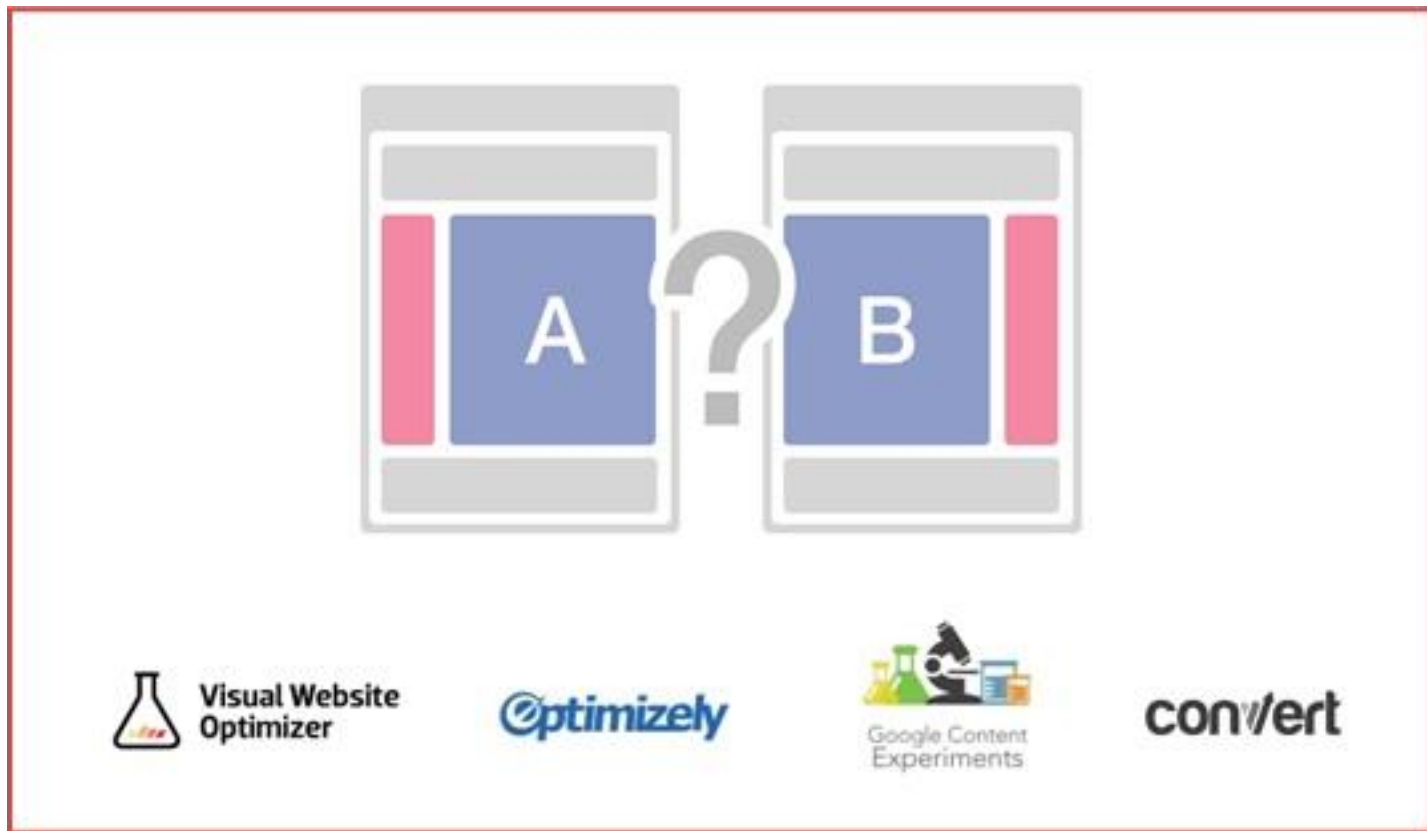
A/B TESTING

- Test 2 variants against each other



TEST TOOLS

- A/B testing: Google Experiments



TEST TOOLS

- Static analyses: Google Analytics

